

Code:

```
#include <stdio.h>
#include <stdlib.h>

#define MAXV 1000005

int count_map[MAXV];
int max_val = 0;
int total_sz = 0;

int main() {
    int N;
    if (scanf("%d", &N) != 1) return 0;

    int i;
    for (i = 0; i < N; i++) {
        int x;
        scanf("%d", &x);
        if (x < MAXV) count_map[x]++;
        if (x > max_val) max_val = x;
        total_sz++;
    }

    int Q;
    if (scanf("%d", &Q) != 1) return 0;

    while (Q--) {
        int val;
        scanf("%d", &val);
        if (val >= MAXV) {
            printf("%d\n", total_sz);
            continue;
        }

        if (val > max_val) {
            max_val = val;
            count_map[val] = 1;
            total_sz++;
        } else if (val == max_val) {
            // max element can only appear once
        } else {
```

```

        if (count_map[val] < 2) {
            count_map[val]++;
            total_sz++;
        }
    }
    printf("%d\n", total_sz);
}

// Print increasing part
for (int v = 1; v <= max_val; v++) {
    if (count_map[v] >= 1) {
        printf("%d ", v);
    }
}
// Print decreasing part
for (int v = max_val - 1; v >= 1; v--) {
    if (count_map[v] == 2) {
        printf("%d ", v);
    }
}
printf("\n");

return 0;
}

```

Input:

```

4
1 2 5 2
4
5
1
3
2

```

Output:

```

4

```

5

6

6

1 2 3 5 2 1